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Assessing Brine Leakage in Saline Sedimentary Basins for Geological Carbon Storage: Pressure Dynamics and the Influence of Faults and Abandoned Wells

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Abstract

We develop a mathematical model to quantify the pressure response due to brine and CO_2 leakage in saline sedimentary basins used for geological carbon storage. The scope includes evaluating the impact of abandoned wells and faults, with a focus on pressure dynamics and hydrological mechanisms that influence leakage pathways. A semi-analytical approach is employed to model fluid migration and leakage in saline sedimentary basins through conductive abandoned wells to the overburden formation. The methodology incorporates the principle of superposition in space to account for the presence of faults and abandoned wells as leaky pathways. Pressure diffusivity equations are used to evaluate leakage rates under varying reservoir and boundary conditions. The model is validated using high-fidelity numerical simulations that models the leakage through abandoned wells in the presence of sealing faults, ensuring robustness and accuracy. The proposed model quantifies the pressure response due to leakage rates with high accuracy, highlighting the significant influence of well integrity and fault characteristics on fluid migration. Observations indicate that poorly sealed abandoned wells contribute significantly to leakage into overlying aquifers. Pressure dynamics reveal critical thresholds at which fluid migration occurs, providing insights for optimal injection site selection and risk mitigation. The results demonstrate the effectiveness of the semi-analytical model in capturing the geological storage mechanisms. The study concludes that addressing abandoned well integrity and fault permeability is essential to ensure the safety and efficacy of geological carbon storage. This study contributes by enhancing the infinite-acting model of injection and abandoned wells through the integration of no-flow boundaries using the method of images. The research presents a novel approach that incorporates sealing faults and abandoned wells as critical leakage pathways into existing semi-analytical models. By providing practical and efficient tools and actionable insights, it enables engineers to better predict and mitigate brine and CO_2 leakage in saline sedimentary basins, thereby enhancing the safety and efficiency of geological carbon storage practices.

Introduction

Sustainable management of greenhouse gases (GHG) has become a vital task for the energy industry and different nations around the globe to achieve net-zero emissions. For instance, carbon emission from Saudi Arabia is about 600 million tons per year, where the country has committed to achieve net zero by 2060 (Hamieh et al., 2022; Rowaihy et al., 2025). One of the promising technologies to reduce GHG emissions and control the temperature rise is Carbon Capture and Storage (CCS). This technique involves capturing CO_2 from the atmosphere and storing it in secure storage sites. Geological carbon storage (CO_2 geo-storage) is recognized as one of the reliable approaches to mitigating CO_2 emissions. Various techniques for underground CO_2 storage include deep ocean CO_2 storage, sequestering CO_2 in depleted gas reservoirs, storing CO_2 in basaltic reactive formations to form stable solid minerals, and storing CO_2 in saline sedimentary basins such as deep saline aquifers (Ali et al., 2023; Oelkers et al., 2022; Omar et al., 2025; Ye et al., 2023). The last technique, relying on structural, solubility, residual, and mineral trapping mechanisms, shows promise due to the global abundance of deep saline aquifers.

There are several hydrological challenges that need to be explored to assess the capability of saline aquifers to contain CO_2. A major hydrological concern for CO_2 geo-storage in saline aquifers is the potential leakage of CO_2 and resident brine through permeable conduits. The oil and gas exploration and production industry has carried out extensive drilling and well-completion activities. However, wells may be abandoned due to inadequate cementing, which can compromise their integrity. Once abandoned, these wells may act as pathways for fluid (brine or CO_2) to migrate or leak from saline aquifers to overlying rock layers, and potentially even to the surface. This leakage mechanism could lead to serious social and environmental implications.

Quantifying the leakage rate and pressure change due to leakage is a challenging task due to insufficient monitoring and geological data. The presented solutions of leakage rate quantification through abandoned wells in the literature could be classified in terms of numerical models or mathematical models. Mathematical models are either analytical or semi-analytical solutions (Nordbotten et al.,2004). Analytical solutions for leakage rates do not require any numerical processing for the final result but provide a closed-form solution without the need for numerical inversion or time-stepping. As a result, they are simple and computationally efficient. In contrast, semi-analytical solutions involve more complexity, often requiring numerical inversion from the frequency domain to the time domain or discretization. These approaches are more intricate with medium computational cost. Conversely, numerical modeling provides a more detailed representation of the system's physics, allowing for the relaxation of the assumptions made in analytical and semi-analytical solutions, despite the cost of increased computational resources.

Most of the existing solutions in the literature focus on an open-domain scenario. This indicates that the domain could be a layered reservoir with injection and abandoned wells, but there are no sealing faults in the system. Subsurface formations are rarely homogeneous in nature, and the existence of no-flow boundaries such as sealing faults can add complexity to the derived solution of the leakage rate and pressure change caused by the leakage. Hence, the motivation behind this work is to improve the open-domain solution for the diffusivity equation by adding faults in the domain using the method of images, which relies on the principle of superposition in time and space. The semi-analytical solution developed in this work can handle CO_2 injection into a subsurface aquifer system with multiple injection wells, multiple layers, multiple abandoned wells, and numerous boundary types such as a channelized system. This paper is organized as follows: Section 2 presents the methodology, followed by a description of the numerical model in Section 3. Section 4 discusses the results, and Section 5 provides concluding remarks.

Methodology

In this research, the pressure response due to the leakage is calculated using the principle of superposition in space and time. Then, the method of images (Lee et al.,2003). is used to account for the existence of the sealing faults. In order to simplify the computations, the methodology starts with a conceptual model that is shown in Fig 1. Then, the mathematical model will be described through a series of equations

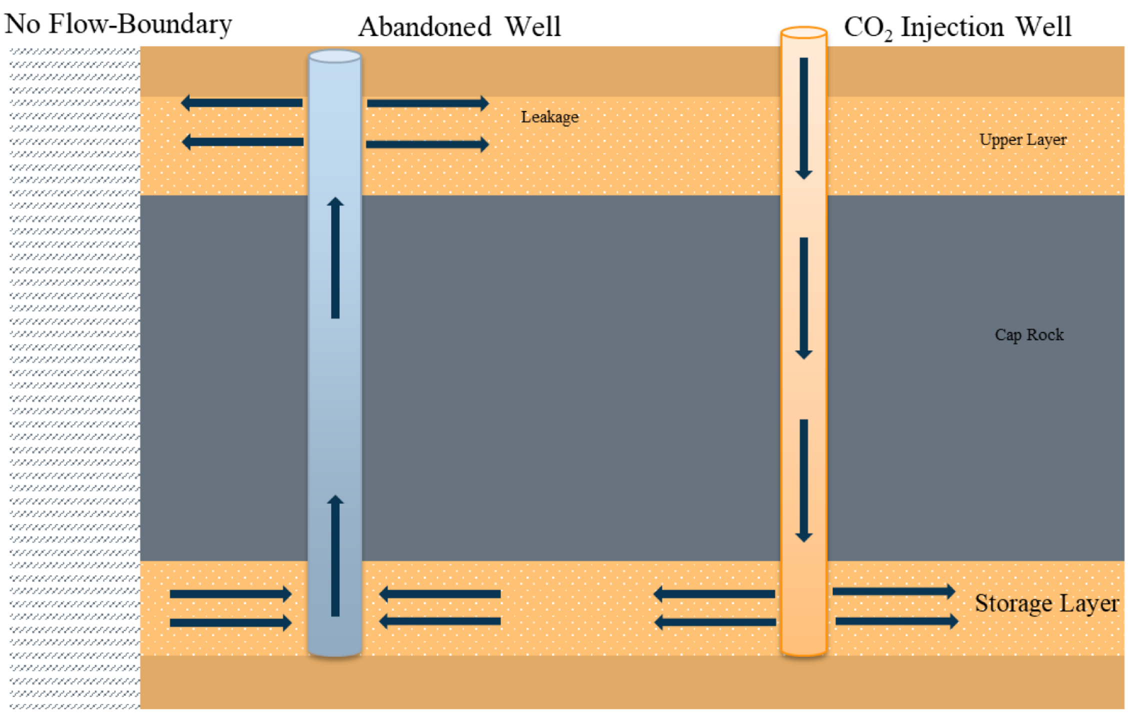

Figure 1—Conceptual illustration of CO_2 injection and leakage in a fault-bounded saline aquifer, showing radial flow from an injection well and upward leakage through an abandoned well, with a lateral no-flow boundary limiting flow beyond the leakage zone.

As shown in Fig. 1, the investigated scenario involves a CO_2 injection well intersecting the caprock (referred to as an aquiclude) of a subsurface reservoir, facilitating fluid injection into the designated storage formation beneath the caprock. An abandoned well is postulated to provide hydraulic connectivity between this storage formation and an overlying geological layer. Both the upper and lower formations are considered to be homogeneous and isotropic and possess uniform thicknesses. A no-flow boundary, represented by a fault, defines the lateral extent of the storage formation. Initially, it is assumed that the injected CO_2 possesses identical physical properties to the resident brine water within both formations, where all fluids are treated as compressible. The abandoned well's leakage pathway is modeled as a permeable conduit with a constant hydraulic resistance. The injection well operates under a constant injection rate, and fluid flow through the porous medium follows Darcy's law, which can be described by the diffusivity equation.

Mathematical Model

The mathematical model for fluid flow in porous media equation is given by the diffusivity equation which is given by Eq. 1:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right)=\frac{\varphi\mu c_t}{k}\frac{\partial p}{\partial t} \tag{1}$$

where p [psi] is the fluid pressure, r [ft] is the radial distance, φ is the medium porosity, k [mD] is the medium permeability, μ [cP] is the fluid viscosity, c_t [psi^{-1}] is the total compressibility, and t [hr] is the time.

To solve Eq. 1, initial and boundary conditions need to be imposed. The initial condition for this problem is given by Eq. 2:

$$p = p_i(t = 0, r \geq 0) \tag{2}$$

where p_i site is the initial reservoir pressure of the storage site, and the gravitational forces were neglected.

The condition at the outer boundary represents an infinite-acting subsurface porous media, in which the pressure remains unchanged. Such a condition could be mathematically expressed as follows:

$$p = p_i(t \geq 0, r \rightarrow \infty) \tag{3}$$

The condition at the inner boundary represents the volume of fluid exchanged between the wellbore and the surrounding subsurface porous media. When a constant injection rate is specified at the injection well with the line-source assumption, the following inner boundary condition is obtained:

$$\lim_{r \rightarrow 0}\left(r\frac{\partial p}{\partial r}\right) = \frac{qB\mu}{2\pi\alpha_2 kh} \tag{4}$$

where q is the injection rate in stock tank barrel per day [STB/D], $\alpha_2 = 0.000264$ is a unit conversion factor. B [rb/STB] is the formation volume factor, and h [ft] is the thickness of the subsurface layers.

The pressure solution of the diffusivity equation at a specific time and location was provided by Thies (1935) for a pumping well. Such a solution could be adapted for injection and leaky wells. Thies's solution is given in Eq. 5 after series of mathematical manipulation, as follows:

$$\Delta p = \frac{qB\mu}{4\pi\alpha_2 kh} E_i\left(-\frac{\varphi\mu c_t r^2}{4\alpha_1 kt}\right) \tag{5}$$

where E_i is the integral exponential function that is defined as follows:

$$E_i(-u) = -\int_u^{\infty}\frac{e^{-u}}{u}du \tag{6}$$

where $u = \frac{\varphi\mu c_t r^2}{4\alpha_1 kt}$ is the argument of the integral exponential function. The exponential integral can be efficiently computed.

In typical CO_2 geo-storage projects, multiple injection wells are used for CO_2 injection and withdrawal wells are employed to produce excess brine, thereby creating additional storage capacity for CO_2 while managing the reservoir pressure. Furthermore, real storage sites often contain several abandoned wells (e.g., abandoned oil and gas producers) through which CO_2 may leak with time-varying leakage rates. The presence of a time-dependent leakage rate necessitates the application of the superposition principle. Consequently, it is essential to apply the superposition principle in both space and time to account for the contributions from all wells present in the storage site. The concept of superposition is illustrated in Fig. 2. Suppose that wells A, B and C are three injection wells with opening time of t_A,t_B and t_C, respectively. Then, Theis's solution for these wells can be written with the superposition in space principle, as follows:

$$\Delta p_{total} = \frac{q_A B\mu}{4\pi\alpha_2 kh} E_i\left(-\frac{\varphi\mu c_t r_{AP}^2}{4\alpha_1 kt_A}\right) + \frac{q_B B\mu}{4\pi\alpha_2 kh} E_i\left(-\frac{\varphi\mu c_t r_{BP}^2}{4\alpha_1 kt_B}\right) + \frac{q_C B\mu}{4\pi\alpha_2 kh} E_i\left(-\frac{\varphi\mu c_t r_{CP}^2}{4\alpha_1 kt_C}\right) \tag{7}$$

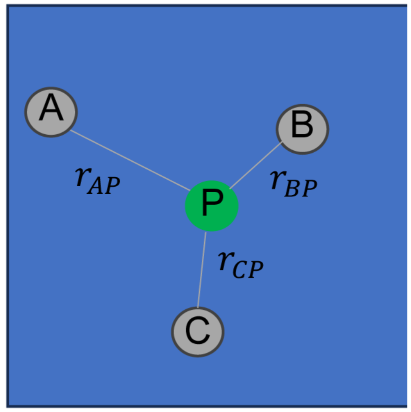

Figure 2—Illustration of the spatial superposition principle in a multi-well system.

The temporal superposition principle is also needed as it is a mathematical technique used to incorporate historical variations in flow rate into analytical models designed for interpreting pressure-transient test data. The leakage through abandoned wells varies with time, necessitating the application of the superposition in time principle. To demonstrate this concept, the time-dependent leakage behavior is represented as a discretized, piecewise-continuous function, as shown in Fig. 3.

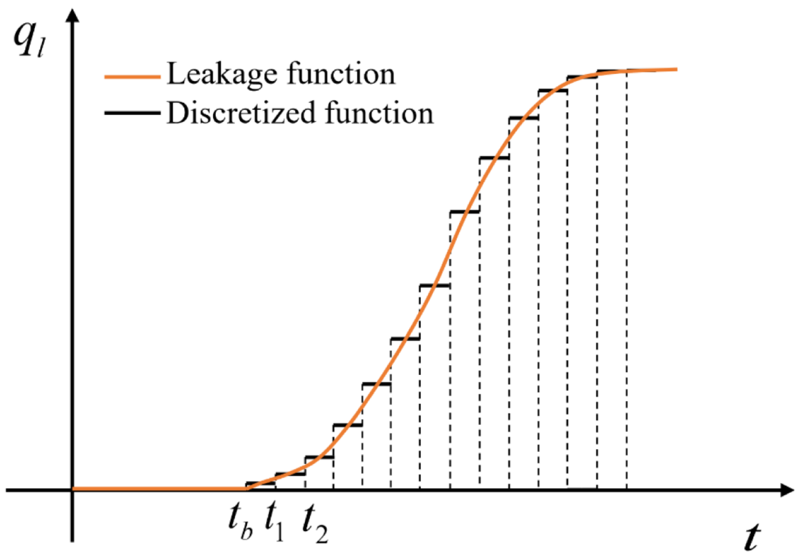

Figure 3—Illustration of a time-dependent leakage function (orange curve) and its corresponding piecewise constant approximation (black stepwise function) (Qiao et al., 2021).

For a multi-well scenario involving variable production rates, consider the configuration shown in Fig. 2. In this case, each well (A, B, and C) may start leaking at a different time and at a variable rate until a common reference time, t_{ref}. The principle of superposition in both space and time can be applied to compute the total pressure drop at point P and is expressed as:

$$\Delta p_{total} = \frac{q_A B\mu}{4\pi\alpha_2 kh} E_i\left(-\frac{\varphi\mu c_t r_{AP}^2}{4\alpha_1 k\left(t_{ref}-t_A\right)}\right) + \frac{\left(q_B-q_A\right)B\mu}{4\pi\alpha_2 kh} E_i\left(-\frac{\varphi\mu c_t r_{BP}^2}{4\alpha_1 k\left(t_{ref}\right)}\right) + \frac{\left(q_C-q_B\right)B\mu}{4\pi\alpha_2 kh} E_i\left(-\frac{\varphi\mu c_t r_{CP}^2}{4\alpha_1 k\left(t_{ref}-t_C\right)}\right) \tag{8}$$

The contribution of this study lies in extending the infinite-acting model involving injection and abandoned wells by incorporating no-flow boundaries through the application of the method of images.

The method of images states that a no-flow boundary, such as a sealing fault, can be represented by an equivalent hypothetical well, termed an image well, operating at the same flow rate and positioned symmetrically on the opposite side of the boundary relative to the real well. As a result, the fault is mathematically eliminated, and the problem is transformed into a system of an infinite-acting reservoir and real and mirrored wells. This concept is depicted in Fig. 4 (a) and (b).

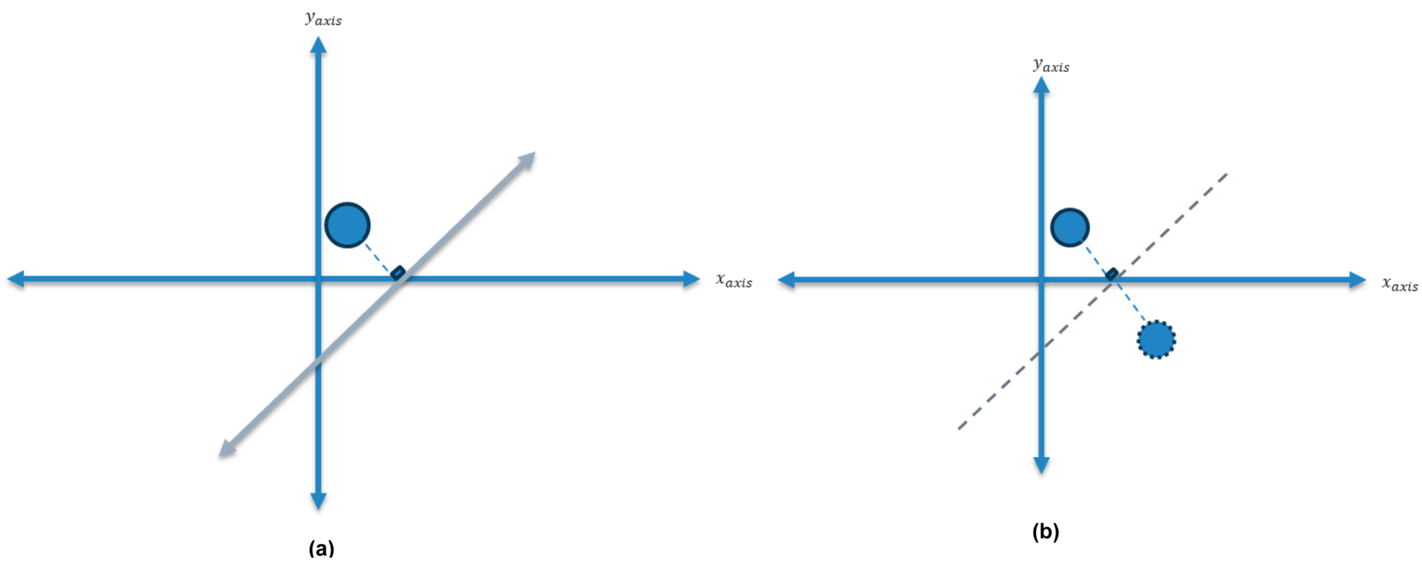

Figure 4—Schematic representation of the method of images. (a)The left part of the figure depicts one actual well with a fault, which acts as a no-flow boundary. (b)In the right part of the figure, the fault is replaced by an image well, positioned symmetrically across the boundary enabling the use of superposition to simplify the analysis.

The approach used in this research to obtain pressure build-up due to the leaky well and under the influence of fault relies on the following principles:

- Theis's solution of the diffusivity equation considering line-source injection well.
- Principle of superposition in space to account for the pressure contribution from the injection and leaky wells.
- Principle of superposition in time to account for the time-varying leakage rate.
- Method of images to consider the sealing fault.

To formulate the solution, a visual perception about the problem statement is provided in Fig. 5. An abandoned well is located at a distance of from the injection well. The influence of the abandoned well on pressure within the storage layer can be accounted for through the application of spatial superposition. By employing this principle, the individual effects of the injection well and the leakage pathway on pressure can be evaluated independently. Moreover, the effect of fault can be considered in the solution by adding imaginary wells that are reflections for the real wells across the fault.

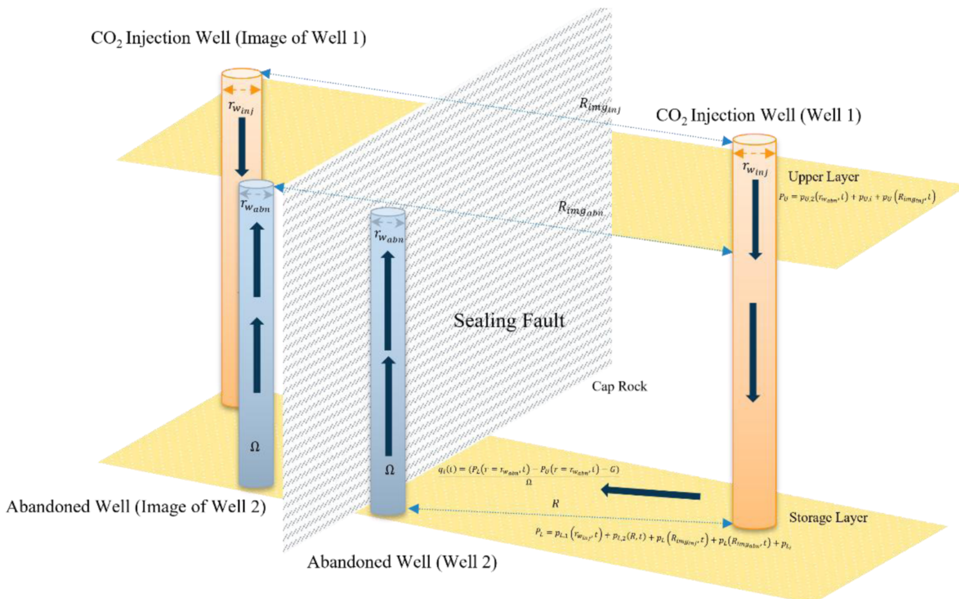

Figure 5—Schematic of a well system using the method of images to model a sealing fault, showing real and mirrored/image injection (orange) and abandoned (blue) wells. Distances between wells R, $R_{img_{inj}}$, $R_{img_{abn}}$ are used to capture pressure interactions and enforce no-flow boundary conditions.

The pressure at the lower layer due to the abandoned well is computed as follows:

$$p_{L,2}(r_{abn},t_n) = a_L(q_n - q_{n-1})E_i\left(b_L\frac{r_{w_{abn}}^2}{t_n - t_{n-1}}\right) + a_L(q_n - q_{n-1})E_i\left(b_L\frac{R_{img_{abn}}^2}{t_n - t_{n-1}}\right) + \ldots$$
$$+ a_L(q_1)E_i\left(b_L\frac{R_{img_{abn}}^2}{t_n - t_b}\right) + a_L(q_1)E_i\left(b_L\frac{r_{w_{abn}}^2}{t_n - t_b}\right) \quad (9)$$

where $R_{img_{abn}}$ $[ft]$ is the distance between image abandoned well and the real injection well, $p_{L,2}(r_{abn}, t_n)$[psi] refers to the bottom hole pressure at the real and imaginary abandoned wells within the lower formation, $r_{w_{abn}}$ represents the abandoned wellbore radius, $a_L\left[\frac{psi}{(STB/D)}\right]$and $b_L\left[\frac{hr}{ft^2}\right]$ are parameters that combine the rock and fluid properties for the lower layer, defined as,

$$a_L = \frac{B\mu}{4\pi\alpha_2 k_L h_L} \quad (10)$$

$$b_L = \frac{\varphi_L \mu c_{t,L}}{4\alpha_1 k_L} \quad (11)$$

The pressure at the upper layer due to the abandoned well is computed as follows:

$$p_{U,2}(r_{w_{abn}}, t_n) = a_U(q_n - q_{n-1})E_i\left(b_U\frac{r_{w_{abn}}^2}{t_n - t_{n-1}}\right) + a_U(q_n - q_{n-1})E_i\left(b_U\frac{R_{img_{abn}}^2}{t_n - t_{n-1}}\right) + \ldots$$
$$+ a_U(q_1)E_i\left(b_U\frac{R_{img_{abn}}^2}{t_n - t_b}\right) + a_U(q_1)E_i\left(b_U\frac{r_{w_{abn}}^2}{t_n - t_b}\right) \quad (12)$$

where $p_{U,2}(r_{w_{abn}}, t_n)$[psi] refers to the bottom hole pressure at the real and imaginary abandoned wells within the upper formation, $a_U\left[\frac{psi}{(STB/D)}\right]$ and $b_U\left[\frac{hr}{ft^2}\right]$ are parameters that combine the rock and fluid properties for the upper layer, defined as,

$$a_U = \frac{\mu}{4\pi\alpha_2 k_U h_U} \tag{13}$$

$$b_U = \frac{\varphi_U \mu c_{t,U}}{4\alpha_1 k_U} \tag{14}$$

The leakage rate is computed by taking the difference between the pressure due to the abandoned well in the lower layer and the upper layer and dividing it by the resistance of the abandoned well to flow. Mathematically, it is written as follows:

$$q_l(t) = \frac{p_{L,2}(rw_{abn}, t_n) - p_{U,2}(rw_{abn}, t_n)}{\Omega} \tag{15}$$

where $q_l(t)$ [STB/D] denotes the leakage rate through the real and image abandoned wells; $r_{w_{abn}}$ [ft] represents the radius of the real and imaginary abandoned wellbores; $P_L(r_{w_{abn}}, t)$[psi] refers to the bottom hole pressure at the real and imaginary abandoned wells within the lower formation; $P_U(r_{w_{abn}}, t)$[psi] signifies the bottom-hole pressure at the same well within the upper formation; and $\Omega\left[\frac{psi}{(STB/D)}\right]$ represents the hydraulic resistance governing flow through the real and imaginary abandoned wellbores.

The computation procedure required to obtain the leakage rate, and the total pressure change due to injection and leakage in the presence of the sealing fault is shown in Fig. 6. The computation process begins by discretizing the total simulation time into a sequence of intervals to facilitate iterative evaluation of leakage rates and pressure responses. Subsequently, the pressure perturbation induced by injection at the abandoned well location is calculated for each time step. For intervals where the pressure perturbation remains zero, the associated leakage rate is set to zero, effectively filtering out inactive periods. The onset of leakage is identified at the breakthrough time, t_b when a non-zero pressure perturbation first occurs; leakage rates are determined using the governing leakage equation, constrained by the abandoned well condition. Finally, the computed leakage rates are superimposed within the pressure equations to derive the overall pressure distribution at the abandoned well.

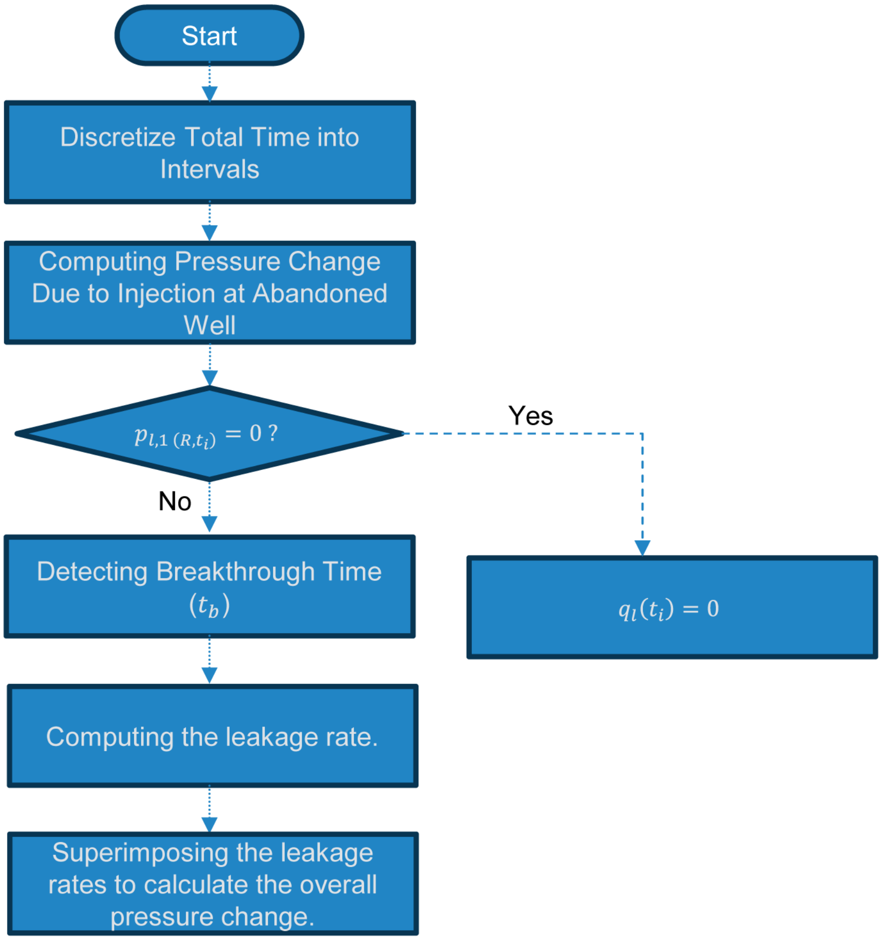

Figure 6—Flowchart outlining the semi-analytical procedure for leakage rate and pressure calculation. The process involves time discretization, pressure computation at the abandoned well, breakthrough detection, leakage rate determination, and superposition of leakage effects to obtain the final pressure distribution.

Numerical Model

Computer Modelling Group (CMG) software is applied in this research. For this study, the IMEX simulator in CMG is adopted to perform numerical analysis. The baseline scenario that is replicated using CMG contains one injection well, one abandoned well, and two active stratified geological units separated by an impermeable layer. This model is depicted in Fig. 7. The pore volumes of the grid cells positioned along the model boundaries were significantly enlarged by several orders of magnitude. This modification ensures that pressure disturbances at the boundaries remain negligible, thereby simulating the conditions of an infinite-acting reservoir. To represent a no-flow boundary associated with the presence of a fault, the volume adjustment at one specific boundary was deactivated, designating that edge as the termination point of the reservoir domain.

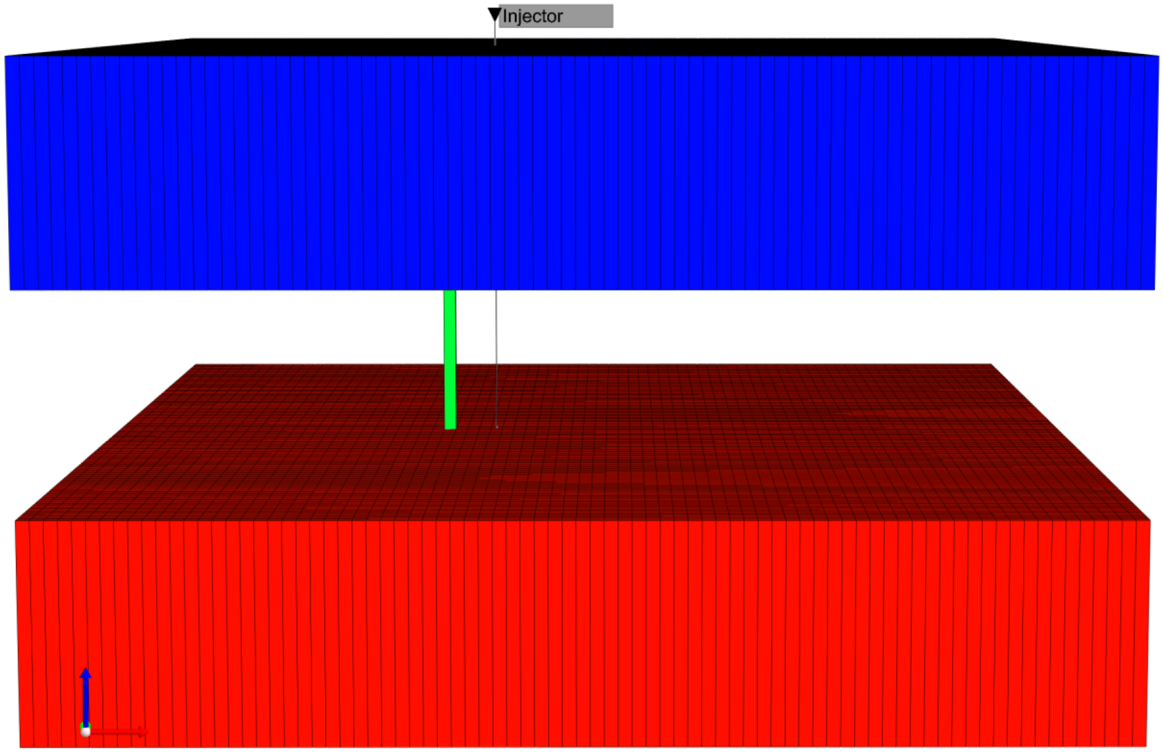

Figure 7—Schematic representation of the numerical simulation model, illustrating the storage layer (red), the overlying impermeable layer (caprock), and the upper geological formation (blue). The model includes an injection well and a vertical conduit (green grid cell) representing an abandoned well, serving as a potential leakage pathway.

The grid cells highlighted in red correspond to the designated storage formation, while the intermediate layer is configured with null blocks, rendering it impermeable and thereby inhibiting fluid transmission. The permeable conduit (green) connecting the storage reservoir to the overlying layer, enabling pressure transmission and fluid migration through this pathway. This configuration effectively simulates the leakage process between the geological formations. The blue grid cells represent the upper layer of the subsurface formation. The specifications of the model are listed in Table 1.

Table 1—Numerical model specifications.

Symbol	Parameter	Value	Unit
h	Formation thickness	50	*ft*
l	Formation length	10100	*ft*
w	Formation width	10100	*ft*
N_x	Number of grids in *x* direction	101	-
N_y	Number of grids in *y* direction	101	-
N_z	Number of grids in *z* direction	10	-
P_i	Initial reservoir pressure	1000	*psi*

Results and Discussion

The physical properties used for the validation of the proposed semi-analytical solution are the same as the properties presented in Table 2. The distance between the injection and abandoned wells is 400 ft. The fault is placed 2900 ft from the leaky well and 3300 ft from the injection well.

Table 2—Reservoir and fluid characteristics utilized in the study.

Symbol	Parameter	Value	Unit
L	Number of layers	3	−
M	Number of injection wells	1	−
N	Number of abandoned wells	1	−
n_f	Number of faults	1	−
FC	Fault coordinates	(2000,0 ,2000, 10100)	*ft*
IWC	Injection well coordinates	(5300, 5100)	*ft*
AWC	Abandoned well coordinates	(4900, 5100)	*ft*
Ω	Resistance of abandoned well to flow	0.1	−
$r_{w_{abn}}$	Abandoned wellbore radius	0.5	*ft*
q	Injection rate	5000	*STB/d*
$r_{w_{inj}}$	Injection wellbore radius	0.5	*ft*
k	Permeability	10	*mD*
h	Thickness	50	*ft*
φ	Porosity	0.3	−
c_t	Total compressibility	$1.2\,e-5$	$1/psi$
μ	Viscosity	1	*cP*
α_2	Unit conversion	0.01416	−
α_1	Unit conversion	0.000264	−

To ensure that the developed semi-analytical model for the brine leakage from the CO_2 storage sites is representative of actual physics, benchmarking with the results obtained from the numerical simulator needs to be undergone. Fig. 8 shows a comparison between the proposed solution of this study and the numerical solution of the bottom-hole pressure at the injection well location in the storage layer. A good match was obtained with the correlation coefficient reaching 0.99.

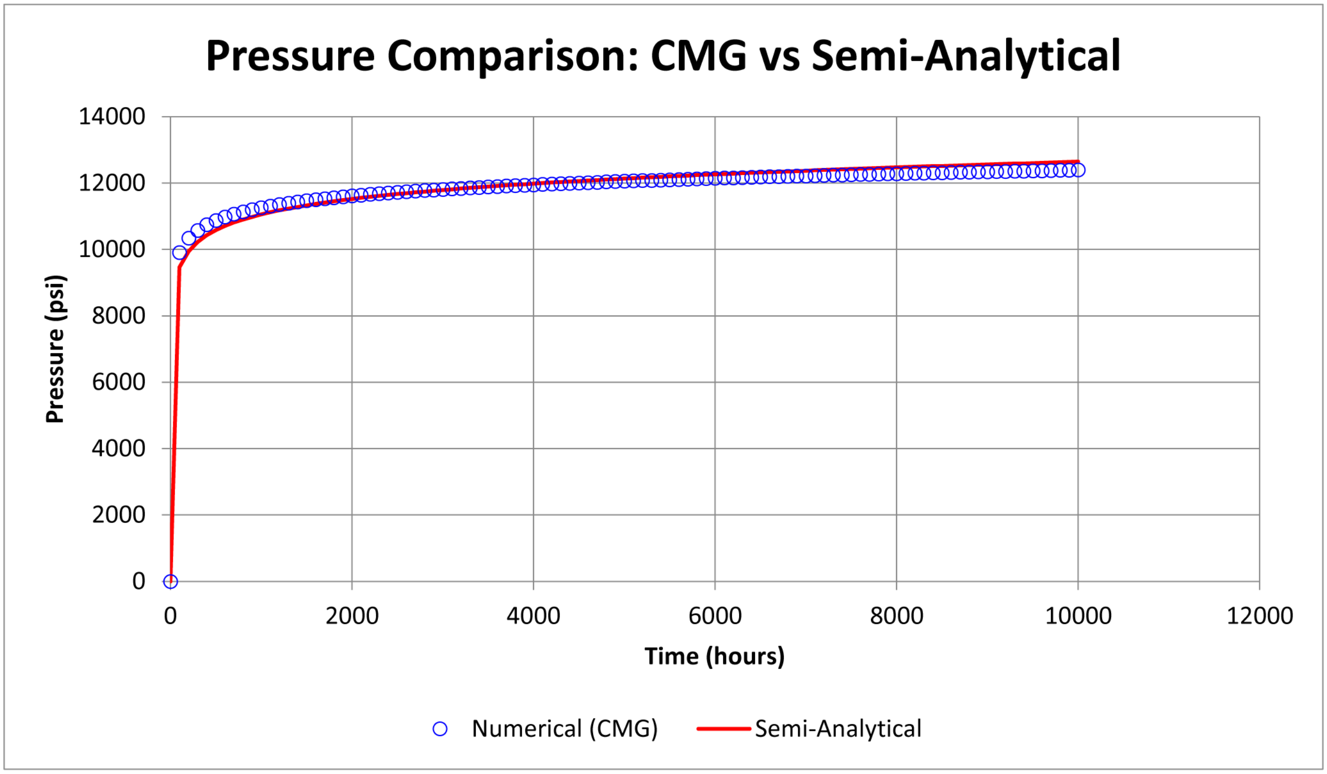

Figure 8—Pressure comparison for the semi-analytical and numerical solution at the injection well location in the storage layer.

Conclusion

This study has focused on CO_2 geo-storage, a key technology for mitigating greenhouse gas emissions and supporting global efforts toward net-zero targets. Among various CO_2 sequestration strategies, storing carbon dioxide in deep saline aquifers remains a promising solution due to their widespread availability and inherent trapping mechanisms. However, the risk of CO_2 and brine leakage through compromised abandoned wells presents a significant hydrological concern, with the potential to impact overlying formations and contaminate freshwater resources. Given the complexity and heterogeneity of subsurface environments, especially in the presence of sealing faults and no-flow boundaries, quantifying leakage rates and pressure variations becomes particularly challenging. To advance the state of knowledge in this domain, this research extended traditional open-domain leakage models by incorporating fault boundaries using the method of images. The developed semi-analytical solution enables the evaluation of pressure changes and leakage rates in more complex geological settings, including systems with multiple layers, injection wells, abandoned wells, and various boundary conditions such as channelized or fully closed domains. Furthermore, the method of image is particularly valuable for its efficiency, as it enables rapid evaluation of leakage rates and pressure buildup in a time-effective manner.

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